

Introduction to COA

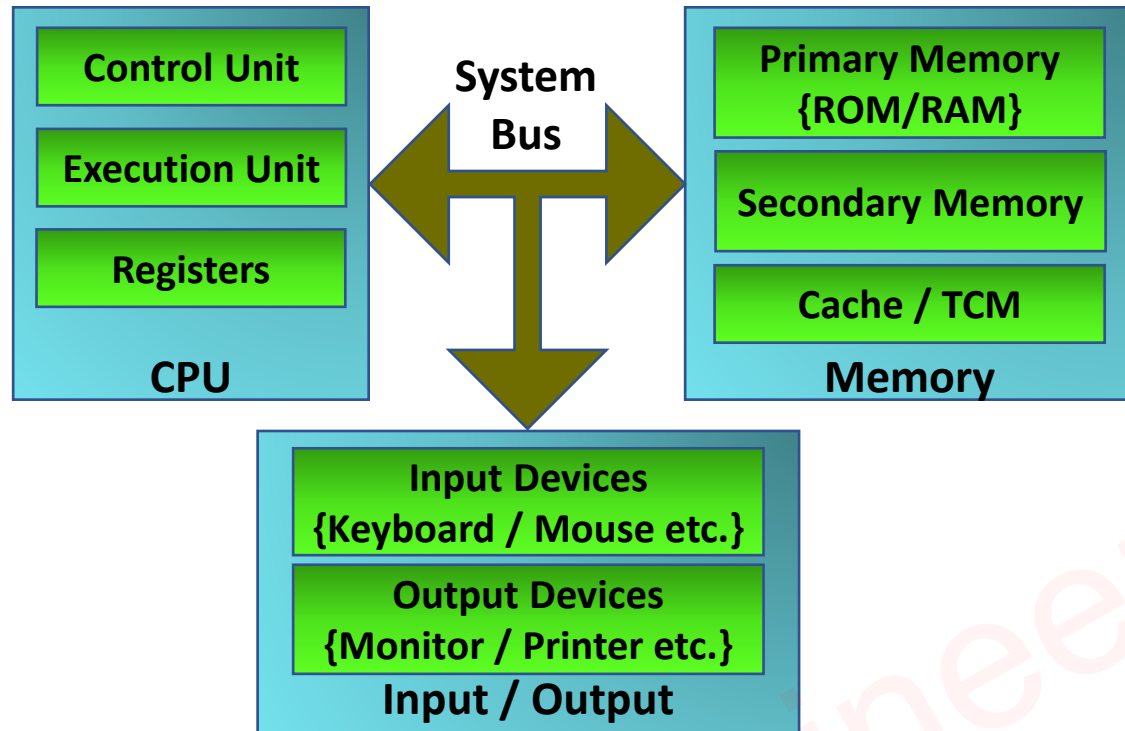
❖ Target Audience of this course

- ❑ CSE/ IT/ EC/ BSC/ MSC – Computer/ IT/ Electronics
- ❑ All the competitive examinations covered:
 - GATE/ BARC/ ISRO/ UGC-NET {8 to 10%}

❖ Books used to prepare this course

- ❑ Morris Mano
- ❑ Pal Choudhary
- ❑ William Stallings
- ❑ John Hayes
- ❑ Andrew Tanenbaum

Computer Organization & Architecture



❖ Memory

- ❑ Memory Hierarchy, Cache Memory & It's Mapping
- ❑ Primary Memory, Secondary Memory
- ❑ TCM, Buffer

❖ Input Output

- ❑ Memory Mapped IO & IO Mapped IO
- ❑ Data Transfer Modes
- ❑ Programmed IO, Interrupt, DMA
- ❑ Serial & Parallel IO
- ❑ Synchronous & Asynchronous IO.

✓ Syllabus

❖ Basics of Computer Architectures

- ❑ Von Neumann & Harvard Architecture
- ❑ Flynn's Classifications, RISC & CISC Architecture
- ❑ Memory calculation
- ❑ Registers organization in computer

❖ Machine Instructions

- ❑ Instructions Formats
- ❑ Instruction cycle, Instructions execution
- ❑ Addressing Modes of Instructions
- ❑ Instruction State transition Diagram

❖ Control Units

- ❑ Hardwire Control Units
- ❑ Microprogrammed Controlled

❖ Execution Unit

- ❑ ALU, Number Representation {2's Complement, IEEE 754}
- ❑ Half & Full Adder, CLA
- ❑ Multiplier {Booth's Algo.}, Division Algorithm {Restoring & Non Restoring methods}

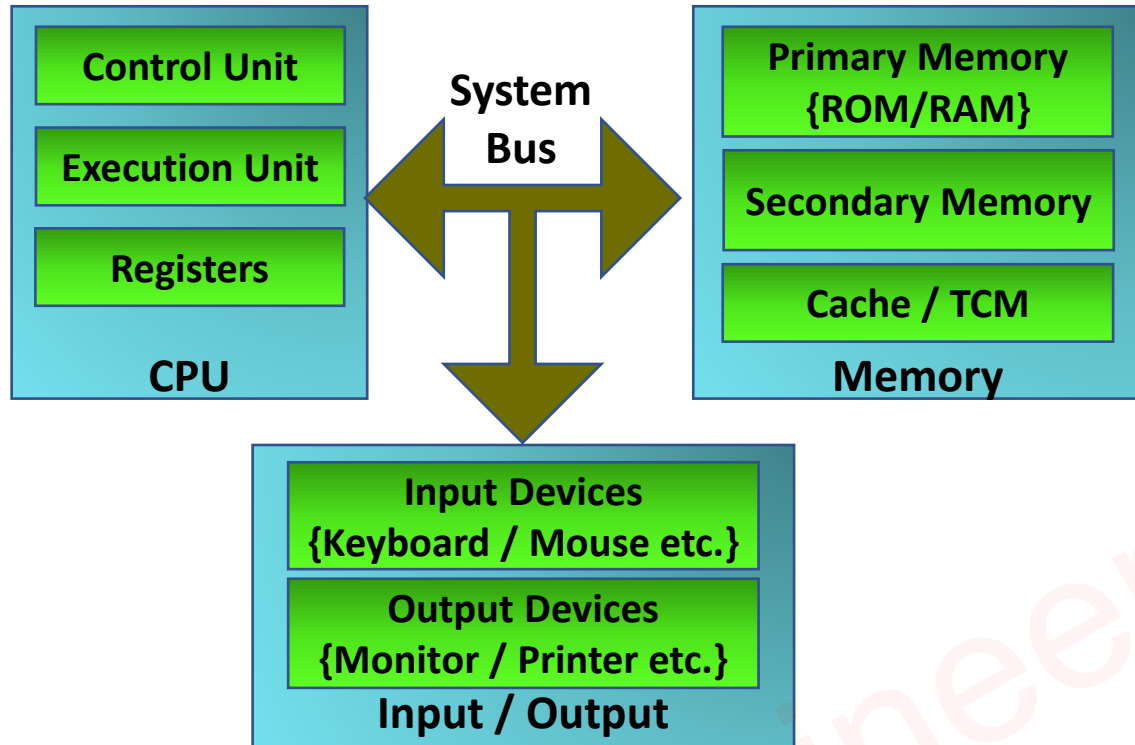
❖ Pipelining

- ❑ Pipelining Process & Pipelining Hazards

❖ System Bus

- ❑ System Bus, Bus Contention and Bus Arbitration Schemes

Computer Architecture



❖ System Bus

- ❑ System bus lines carries information.
- ❑ System includes Address bus, Data bus and control bus.
- ❑ Address bus: it carries address information. It explains how much physical memory can be interfaced on lines.
- ❑ Data bus: it carries data information. It explains how much data can be exchanged in single machine cycle.
- ❑ Control bus; it carries control information. It explains which operation should be performed. {Read/ Write}

❖ CPU – Central Processing Unit

- ❑ It is Heart of computer system.
- ❑ It consists Control Unit, Execution Unit & Registers.
- ❑ It executes all the programs of computer.
- ❑ Program execution is execution of instructions in sequence.
- ❑ Instruction execution happens in three stages:
 - Fetch
 - Decode
 - Execute
- ❑ Fetch and Decode is performed by control unit & execute is performed by ALU.

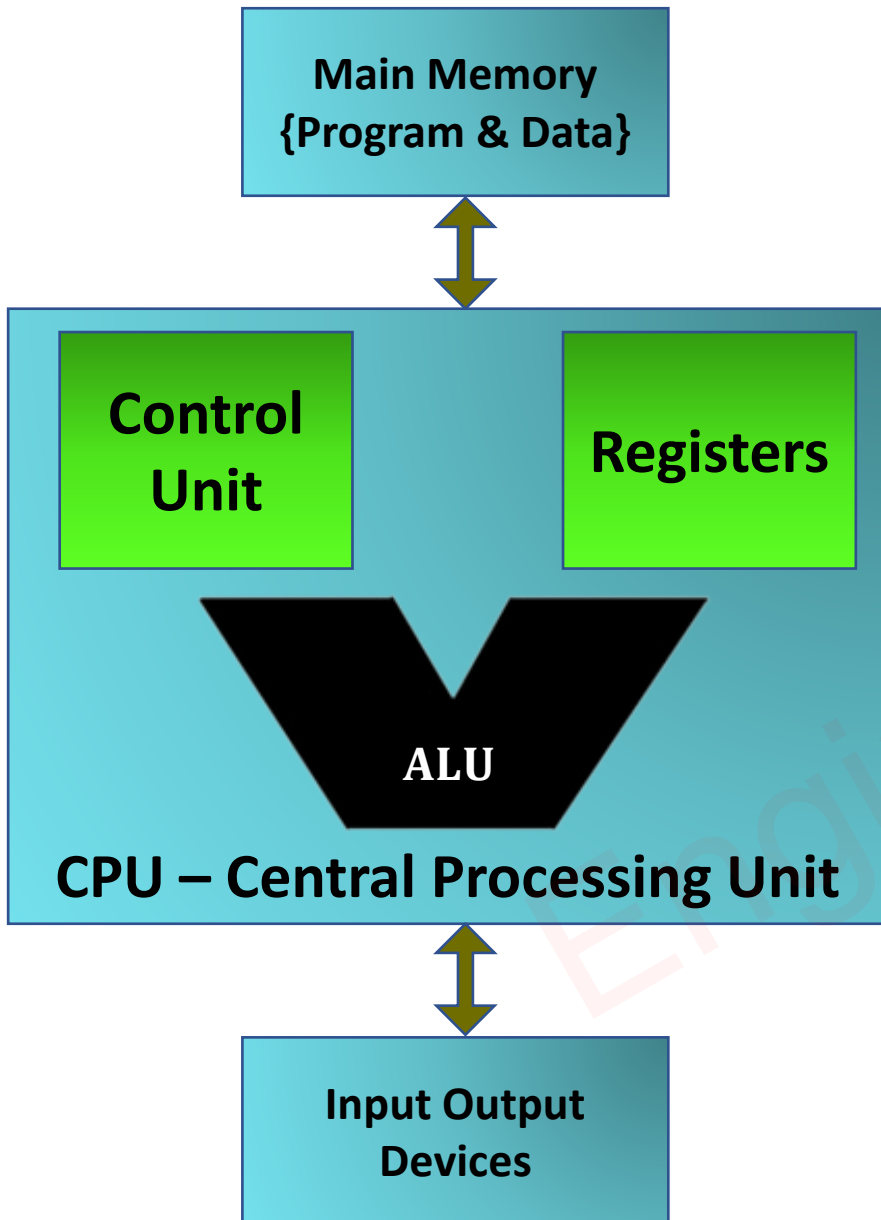
❖ Memory Unit

- ❑ Memory stores programs and data.
- ❑ Each instruction and data is stored at unique location into memory.
- ❑ Primary Memories are ROM and RAM.
- ❑ Secondary Memories are like Floppy Disk, Hard disk etc.
- ❑ Cache Memory is high speed memory built using SRAM.

❖ IO devices

- ❑ It is used to perform Input and Output operations.
- ❑ Input device can be keyboard, Mouse etc.
- ❑ Output device can be Monitor, printer etc.
- ❑ Touch Screen can be used as Input and Output.
- ❑ IO devices are used to access computer system.

Von Neumann Architecture



```
Void main ()
```

```
{  
  Int A = 8, B = 5, C;  
  C = B + A;  
}
```

Higher Level language

Compiler

```
0001001001000101
```

```
0000100110101001
```

```
0101010010010100
```

```
1000100100110010
```

```
1010101110110110
```

Machine language {Program & data}

❖ Main Memory of Von Neumann Architecture

- ❑ It consists program & data in same memory. So common system bus is there for program & data access.

❖ Control Unit

- ❑ It provides timing control, instruction fetch, instruction decode, pipelining, Memory Access etc.

❖ Registers

- ❑ It holds data and it can be operand of ALU.
- ❑ It is very fast in accessing data with memory.
- ❑ Example: General Purpose registers, Flag Register, Stack Register, Program Counter etc.

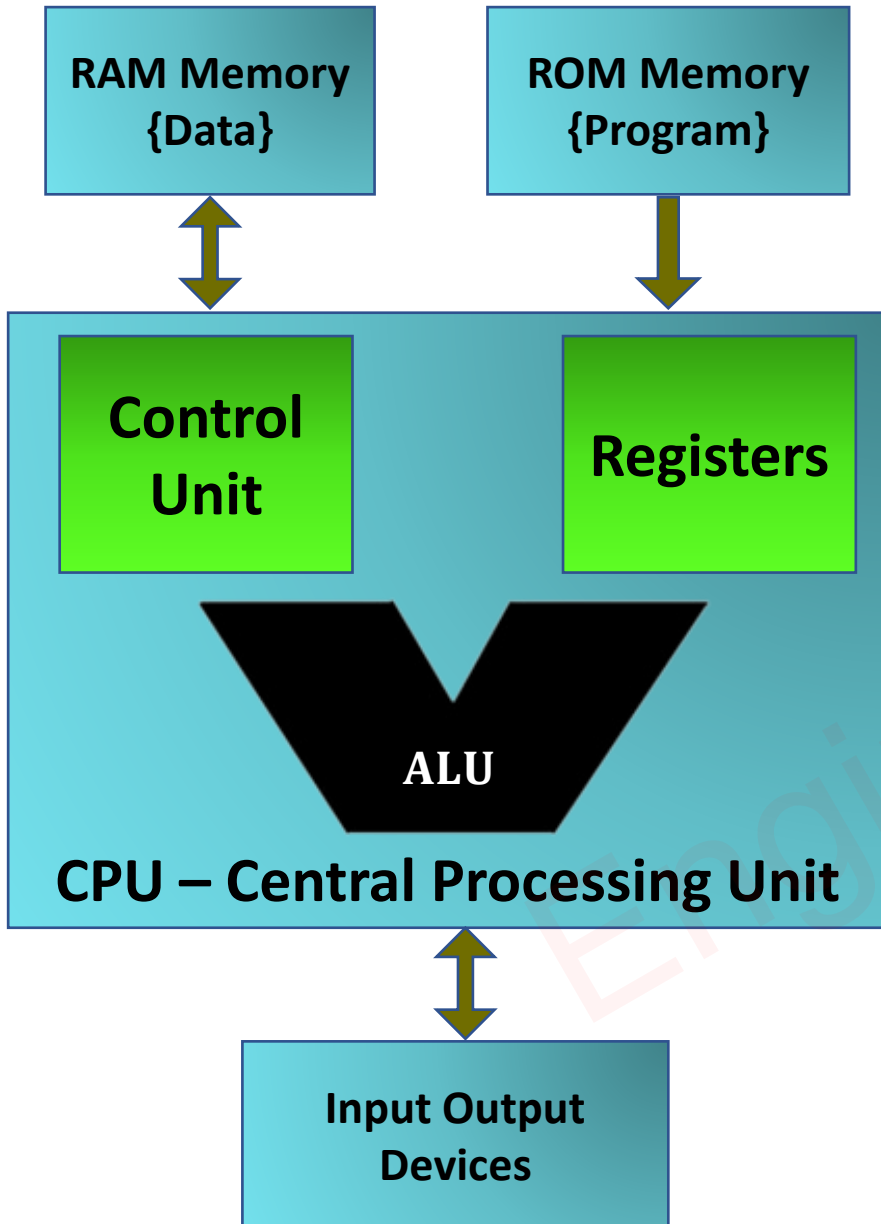
❖ ALU

- ❑ It performs all the arithmetic and logical tasks. {+, -, *, /, ^, |}

❖ IO devices

- ❑ It used for user interface with computer system for input & Output.

Harvard Architecture



```
Void main ()  
{  
  Int A = 8, B = 5, C;  
  C = B + A;  
}
```

Higher Level language



```
000100100 001100100  
100010100 101010100  
001001101 101001101  
010010101 110011101  
010010010 011110010
```

Machine language {Program & data in separate memory}

- ❖ Main Memory of Harvard Architecture
 - ❑ It consists program & data in separate memory. So program {Instructions} is stored in ROM and Data is stored in RAM memory.
- ❖ Control Unit
 - ❑ It provides timing control, instruction fetch, instruction decode, pipelining, Memory Access etc.
- ❖ Registers
 - ❑ It holds data and it can be operand of ALU.
 - ❑ It is very fast in accessing data with memory.
 - ❑ Example: General Purpose registers, Flag Register, Stack Register, Program Counter etc.
- ❖ ALU
 - ❑ It performs all the arithmetic and logical tasks. {+, -, *, /, ^, |}
- ❖ IO devices
 - ❑ It used for user interface with computer system for input & Output.

Von Neumann Vs Harvard



Parameters	Von Neumann	Harvard
Memory	❖ Data and Program {Code} are stored in same memory	❖ Data and Program {Code} are stored in different memory
Memory Type	❖ It has only RAM for Data & Code	❖ It has RAM for Data and ROM for Code
Buses	❖ Common bus for Address & Data/Code	❖ Separate Bus Address & Data/Code
Program Execution	❖ Code is executed serially and takes more cycles	❖ Code is executed in parallel with data so it takes less cycles.
Data/Code Transfer	❖ Data or Code in one cycle	❖ Data and Code in One cycle
Control Signals	❖ Less	❖ More
Space	❖ It needs less Space	❖ It needs more space
Cost	❖ Less	❖ Costly

Flynn's Classifications

❖ Flynn's Classifications

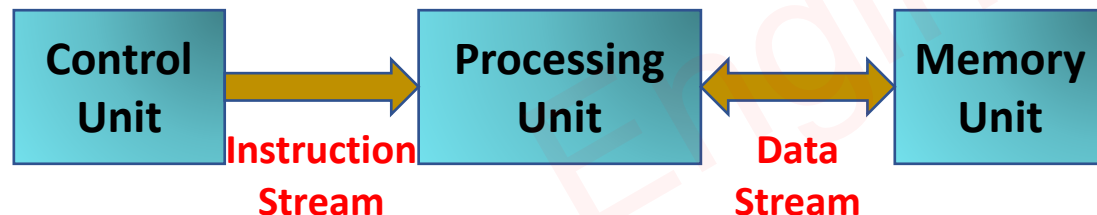
- ❑ It is a way of organizing multiple processor system.
- ❑ Flynn has explained the most common approach for categorizing the systems with parallel processing capabilities.

❖ Categories of Computer systems

- ❑ SISD – Single Instruction Single Data stream
- ❑ SIMD – Single Instruction Multiple Data stream
- ❑ MISD – Multiple Instruction Single Data stream
- ❑ MIMD – Multiple Instruction Multiple Data stream

❖ SISD – Single Instruction Single Data stream

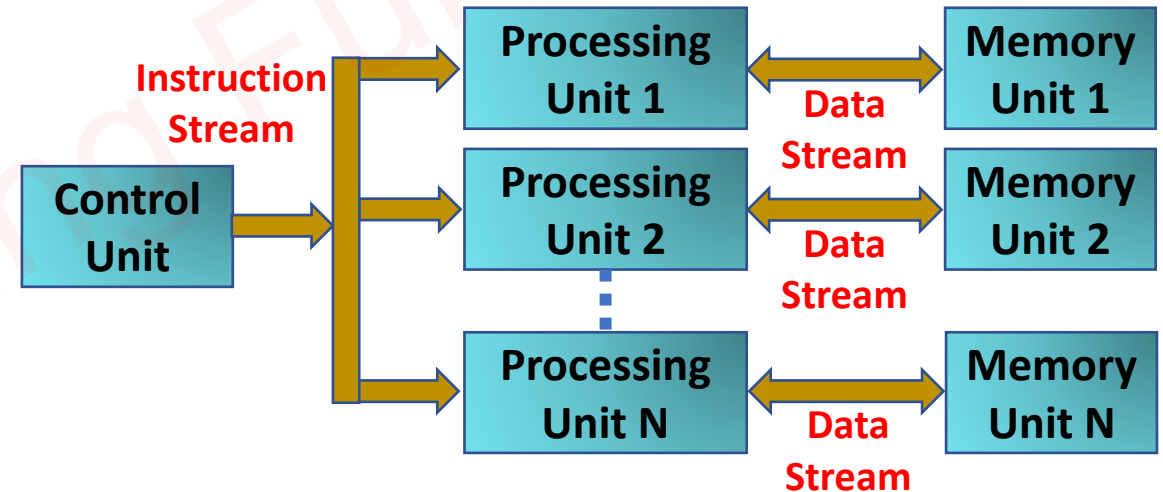
- ❑ In SISD, one instruction runs on one processor on the data stored on single memory.
- ❑ Uniprocessor comes in this category.



- ❑ Machine Instructions are executed in sequential order.
- ❑ This architecture is also known sequential computers.
- ❑ Many of conventional computer follows this architecture.
- ❑ Instructions and data must be stored in primary memory.
- ❑ Clock defines the speed of processing in SISD.

❖ SIMD – Single Instruction Multiple Data stream

- ❑ In SIMD, one instruction controls multiple simultaneous execution of number of processing elements on lock step basis.
- ❑ Each Processing unit has associate data memory for execution of instruction.



- ❑ Scientific computing based on SIMD Architecture.
- ❑ Example of this model are vectors and arrays.

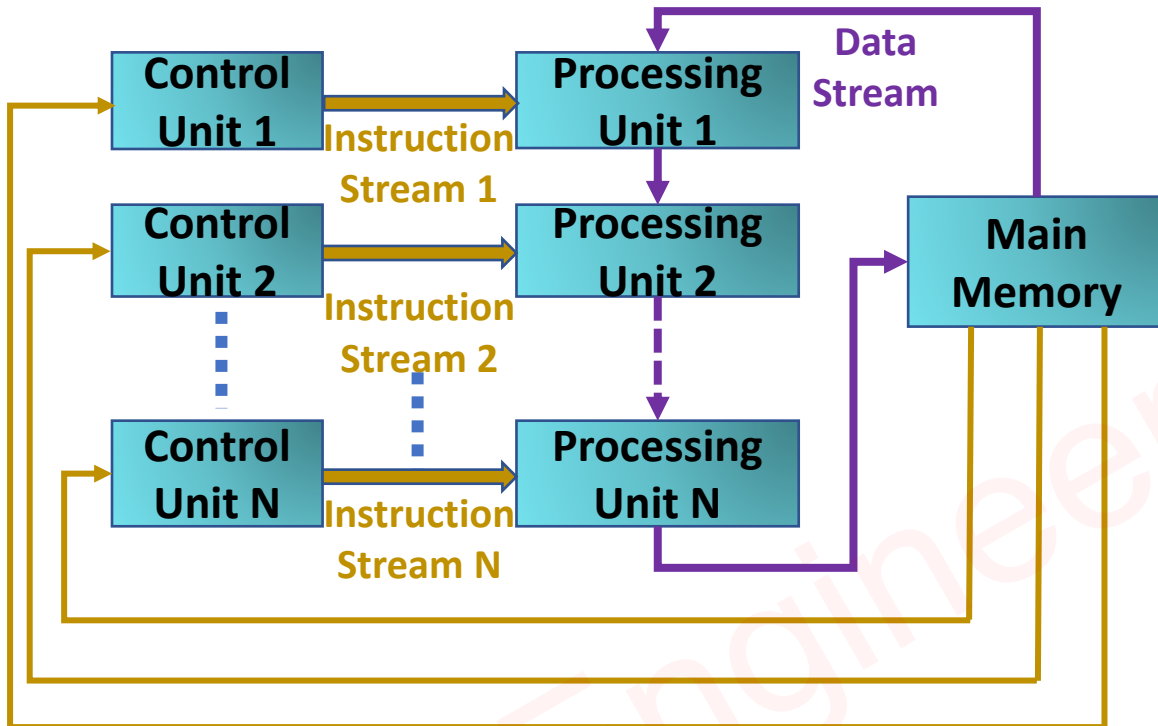


- ❑ If we want to perform arithmetic operation with all the element of Array A then it can be done by this architecture.
- ❑ Example: Add 10 with all elements, Multiply 5 with all element, rotate 3 bits right of all the elements.

Flynn's Classifications

❖ MISD – Multiple Instruction Single Data stream

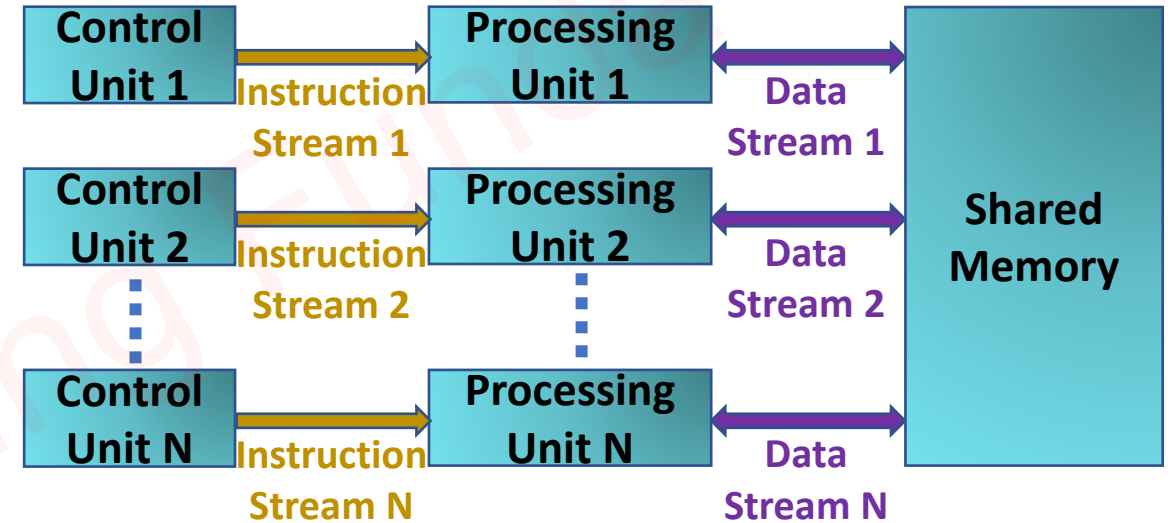
- ❑ In MISD, Here multiple processor executes different instructions with single data stream.



- ❑ This architecture is not implemented commercially.
- ❑ Example : Lets have same data Δ to be used in multiple instructions.
 - $A = \sin \Delta$
 - $B = \cos \Delta$
 - $C = \tan \Delta$

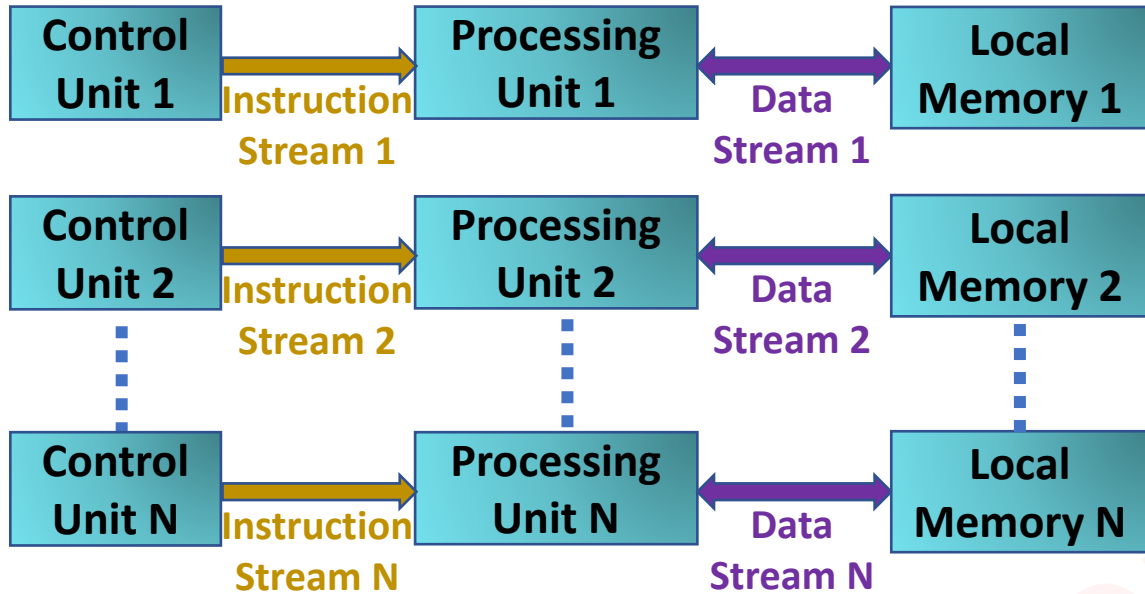
❖ MIMD – Multiple Instruction Multiple Data stream

- ❑ In MIMD, Here multiple processor executes different instructions with multiple data stream.

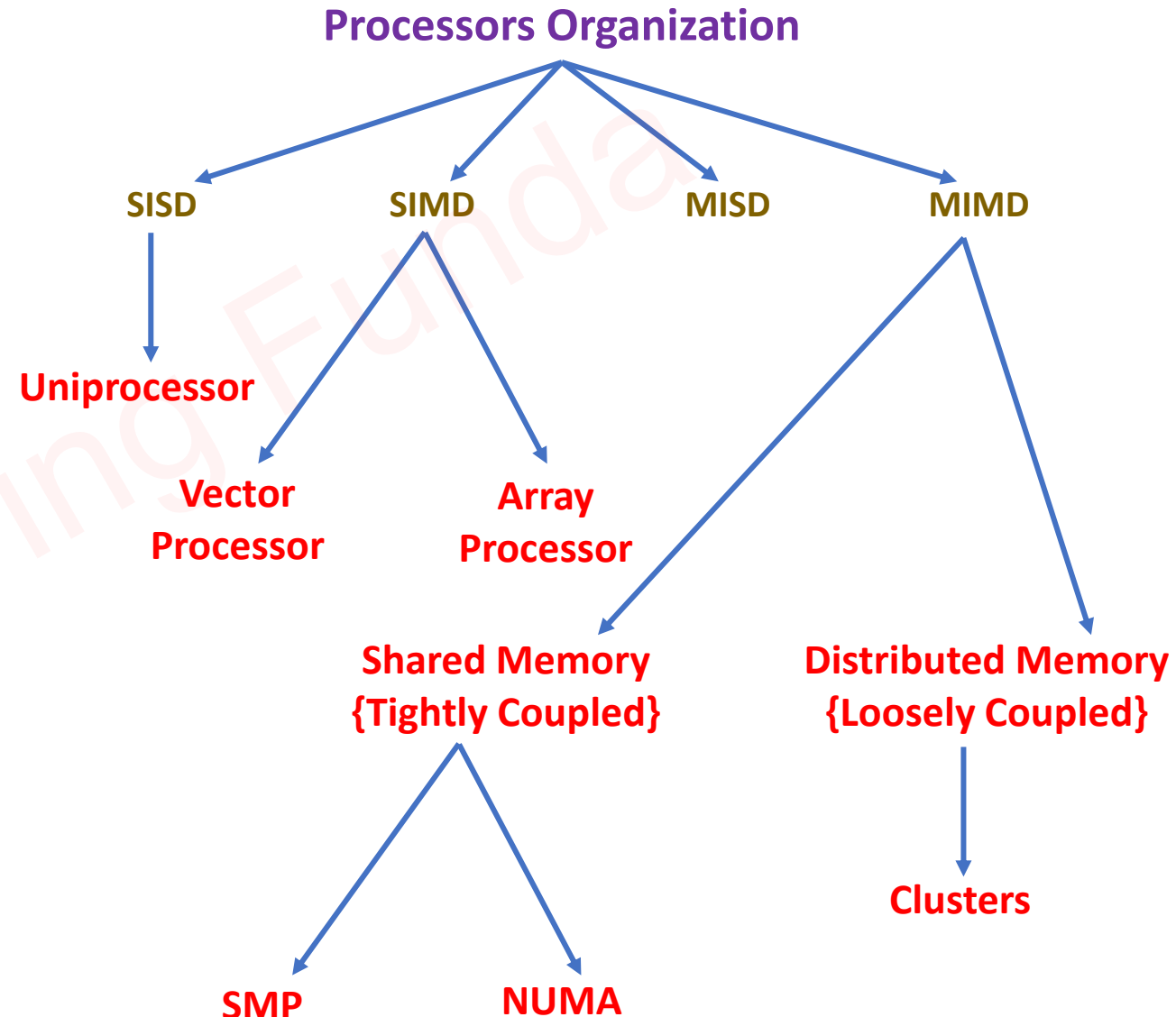


- ❑ Shared Memory in MIMD system is Symmetric multiprocessor system (SMP).
- ❑ Different processor takes instructions and data from common shared memory.
- ❑ With the use of buses, multiple processor executes program along with shared memory.

Flynn's Classifications



- ❑ Local Memory with MIMD is Non Uniform Memory Access NUMA organization.
- ❑ NUMA is costlier compared to SMP.
- ❑ Collection of Uniprocessor or SMP can be used to form cluster.
- ❖ **Symmetric Multiprocessor SMP characteristics**
 - ❑ There are two or more similar or comparable processor.
 - ❑ Memory Access time for all the processor will be same.
 - ❑ All the processors shares IO devices through same channel.
 - ❑ All the processors can perform same function. {That's why it is called symmetric multiprocessor system}



RISC Vs CISC

Parameters	RISC	CISC
Full Form	❖ Reduced Instruction Set Computer	❖ Complex Instruction Set Computer
Instruction Size	❖ Fixed Size	❖ Variable Size
Instruction Fetch Time	❖ Same for all instruction	❖ Vary with respect to instructions
Instruction Set	❖ Small & Simple	❖ Large and Complex
Addressing Modes	❖ Less Modes as most instructions are based on registers	❖ More Modes as Complex instructions are available with different verities.
Numbers of Registers	❖ Many	❖ Few
Complier Design	❖ Simple	❖ Complex
Program Size	❖ Long {Weak code density}	❖ Small {Better Code density}
Numbers of Operand	❖ Fixed {Mainly in Registers}	❖ Variable {Can be in Registers & Memory}
Control Unit	❖ Hardwire controlled	❖ Micro Program Controlled
Execution Speed	❖ Faster	❖ Slower
Pipelining	❖ More Effective	❖ Less Effective {It has more bubbles due more memory based instructions}
Processor	❖ More Suitable for dedicated operations.	❖ More suitable for verities of operations.

Little Endian & Big Endian

- ❑ There are two ways of ordering the data while storing it into the Memory.
 - **Big Endian**
 - **Little Endian**
- ❑ **Big Endian** : Most Significant Byte stored 1st in the memory and Last significant Byte stored at last in the memory.
- ❑ **Little Endian** : Last Significant Byte stored 1st in the memory and Most significant Byte stored at last in the memory.
- ❑ Suppose we want to store 12345678H data starting at address 1000H.

Add	Data
1000H	12H
1001H	34H
1002H	56H
1003H	78H
1004H	

Big Endian

Add	Data
1000H	78H
1001H	56H
1002H	34H
1003H	12H
1004H	

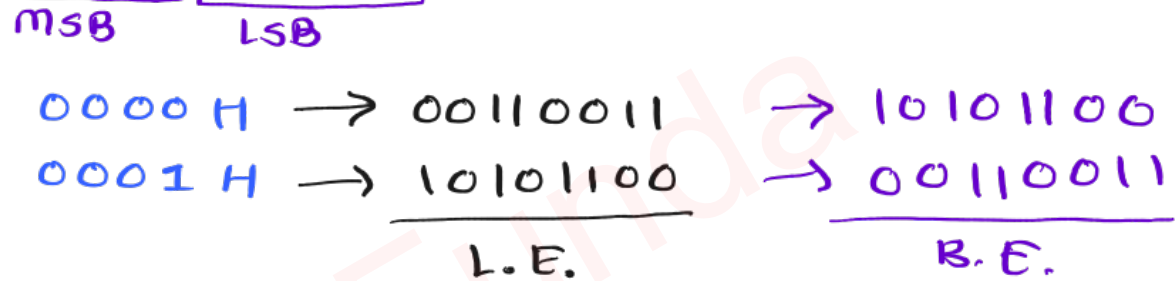
Little Endian

- ❑ As we also operate with stack memory in CPU, we can remember this as per:
- ❑ In Big Endian, Lower Address holds higher Byte of data and Higher Address holds lower Byte of data.
- ❑ In Little Endian, Lower Address holds Lower Byte of data and Higher Address holds Higher Byte of data.
- ❑ Big Endian examples: Motorola 68xx, IBM Mainframe, TCP/IP etc.
- ❑ Little Endian examples: Intel x86, AMD, PIC etc.
- ❑ Advantages of Big Endian:
 1. Easier to determine sign
 2. Easier to compare two number
 3. Easier to divide two number
 4. Easier to Print
- ❑ Advantages of Little Endian:
 1. Easier for Multiplication & Addition of multi-precision numbers.

Examples on Little Endian & Big Endian

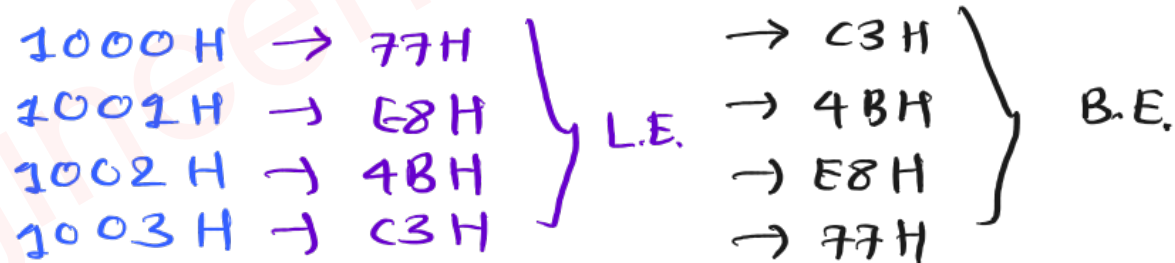
- ☐ **GATE CS** – We are given two bytes data as $10101100\ 00110011$. How it will be stored in Little Endian format?

- ☒ **B.E.** A. $0x0000 = 10101100, 0x0001 = 00110011$
B. $0x0000 = 00110011, 0x0001 = 00110011$
C. $0x0000 = 10101100, 0x0001 = 10101100$
☒ **L.E.** D. $0x0000 = 00110011, 0x0001 = 10101100$



- ☐ **GATE CS** – A Hexadecimal number $C34BE877H$ is stored from memory location $1000H$ in little Endian Format, What will be the value occupied by memory locations $1000H, 1001H, 1002H$ & $1003H$?

- ☒ **B.E.** A. $C3H, 4BH, E8H$ & $77H$
B. $C3H, E8H, 4BH$ & $77H$
☒ **L.E.** C. $77H, E8H, 4BH$ & $C3H$
D. $77H, 4BH, E8H$ & $C3H$



- ☐ **ISRO 2007 CS** – In the Big Endian system, the computer stores

- ☒ A. MSB of Data in the lowest memory address of data unit
B. LSB of Data in the lowest memory address of data unit
C. MSB of Data in the Highest memory address of data unit
☒ D. LSB of Data in the Highest memory address of data unit

GATE Questions of Computer Organization

☐ **GATE CS** – Among the given options choose correct one:

- I. Von Neumann Architecture shares common memory for instructions and data.
- II. Harvard Architecture has separate memory for instructions and data.

- A. Only I is true
✓ B. I and II are true
C. Only II is true
D. I and II are false

☐ **GATE CS** – Suppose we want to compute the value of y^6 , $\sqrt{y}$, $y^{1/3}$ using the value of y . which of the following among Flynn's CPU classification will be best suited for above calculation?

- A. SISD Stream
B. SIMD Stream
✓ C. MISD Stream
D. MIMD Stream

multiple instructions single data

☐ **GATE CS** – Vector and Arrays belongs to which class of Flynn's Classification.

- A. SISD Stream
✓ B. SIMD Stream
C. MISD Stream
D. MIMD Stream

multiple data

$$J = (J_0, J_1, J_2, J_3, \dots, J_n)$$

$J \times 2 \leftarrow$ Single Instruction

Memory Interfacing Questions of COA

- ❑ **GATE CS** – If there are 512G locations in the memory, then find the number of bits to represent each location.

$$\begin{aligned}\Rightarrow \text{No of locations} &= 512 \text{ G} \\ &= 2^9 \cdot 2^{30} \\ &= 2^{39} = 2^n\end{aligned}$$

$$\Rightarrow n = 39 \text{ bits}$$

- ❑ **GATE CS** – If there are 26 bits available to represent each word of memory, then how many locations can be addressed uniquely?

$$\Rightarrow n = 26 \text{ bits}$$

$$\begin{aligned}\Rightarrow \text{No. of locations (Address)} &= 2^n \\ &= 2^{26} \\ &= 2^6 \times 2^{20} \\ &= 64 \text{ M}\end{aligned}$$

□ **GATE CS** – If memory is byte addressable and size of memory is 512KB. Calculate the number of bits required to represent each word.

↑
with each
address we have
one byte data

↑ ↑
Address data

$$\begin{aligned}\Rightarrow \text{No of locations \{ Address \}} &= \frac{\text{Size of Memory}}{\text{Size of each location}} \\ &= \frac{512 \text{ KB}}{B} \\ &= 512 \text{ K} \\ &= 2^9 \times 2^{10} \\ &= 2^{19}\end{aligned}$$

$\Rightarrow n = 19$ bits.

- ❑ **GATE CS** – If the size of memory is 512GB. Size of word is 16bits. Find the numbers of bits required to represent each word uniquely. Also Find the number of locations required to represent each bit uniquely.

$$\begin{aligned}\Rightarrow \text{No of Locations} &= \frac{\text{Size of Memory}}{\text{Size of Location}} \\ &= \frac{512 \text{ GB}}{2} \\ &= 256 \text{ G} \\ &= 2^8 \times 2^{30} \\ &= 2^{38}\end{aligned}$$

$$\Rightarrow n = 38 \text{ bits.}$$

$$\begin{aligned}\Rightarrow \text{Number of locations} &= \frac{\text{Size of Memory}}{\text{Size of Location}} \\ &= \frac{512 \text{ GB}}{1} \\ &= 2^9 \times 2^{30} \times 2^3 \\ &= 2^{42}\end{aligned}$$

$$\Rightarrow n = 42 \text{ bits.}$$

❑ **GATE CS** – If memory size is 64GB. If word size is 32bits. Then find the memory capacity in Words, Bytes and bits. Also find the numbers of bits required to represent words, bytes and bits.

$$\Rightarrow \text{No of locations \{ Addresses \}}_{\text{word}} = \frac{\text{Size of Memory}}{\text{Size of word}} = \frac{64 \text{ GB}}{32} = \frac{64 \times 2^6 \times 2^{30} \times 8}{32} = 2^{\textcircled{34}} \\ = 2^4 \times 2^{30} = 16 \text{ G}$$

$\Rightarrow n_{\text{word}} = 34 \text{ bits.}$

$$\Rightarrow \text{No of locations \{ Addresses \}}_{\text{B}} = \frac{\text{Size of Memory}}{\text{Size of word}} = \frac{64 \text{ GB}}{8} = 64 \text{ G} = 2^6 \times 2^{30} \\ = 2^{\textcircled{36}}$$

$\Rightarrow n_{\text{byte}} = 36 \text{ bits.}$

$$\Rightarrow \text{No of locations \{ Addresses \}}_{\text{bits}} = \frac{\text{Size of Memory}}{\text{Size of word}} = \frac{64 \text{ GB}}{1} = 64 \times 2^6 \times 2^{30} \times 2^3 \\ = 2^6 \times 2^{30} \times 2^3 = 2^{\textcircled{39}} = 512 \text{ G}$$

$\Rightarrow n_{\text{bits}} = 39 \text{ bits.}$

Memory Interfacing Questions of COA

❑ **GATE CS** – How many 256MB RAM chips required to build 4GB RAM?

$$\begin{aligned}\Rightarrow \text{Number of chips} &= \frac{\text{Total Size}}{\text{Available Size}} \\ &= \frac{4\text{GB}}{256\text{ MB}} = \frac{2^2 \times 2^{30} \times \cancel{2^3}}{2^8 \times 2^{20} \times \cancel{2^3}} = 2^4 = 16 \text{ chips}\end{aligned}$$

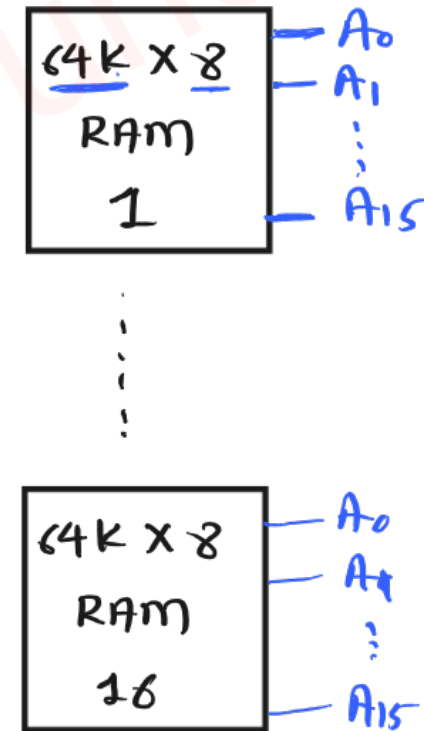
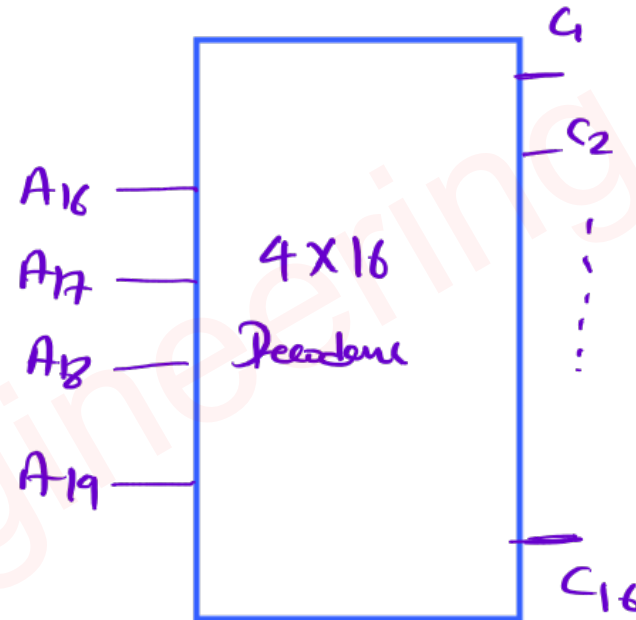
❑ **GATE CS** – How many 128KB RAM chips required to build 1MB RAM?

$$\begin{aligned}\Rightarrow \text{Number of chips} &= \frac{\text{Total Size}}{\text{Available Size}} \\ &= \frac{1\text{MB}}{128\text{ KB}} = \frac{2^{20} \times \cancel{2^3}}{2^7 \times 2^{10} \times \cancel{2^3}} = 2^3 = 8 \text{ chips}\end{aligned}$$

❑ **GATE CS** – How many 64K X 8 RAM chips are required to build 1MB of RAM. Also Calculate the size of decoder.

$$\Rightarrow \text{Number of Chips} = \frac{\text{Total Size}}{\text{Available Size}} = \frac{1 \text{ MB}}{64 \text{ K} \times 8} = \frac{2^{20} \times 2^3}{2^5 \times 2^{10} \times 2^3} = 2^4 = \underline{16 \text{ Chips}}$$

$$\Rightarrow \begin{array}{c} \boxed{\begin{array}{cc} 4 & \times & 16 \\ \hline \uparrow & & \uparrow \\ n & & 2^n \end{array}} \leftarrow \text{Size of decoder.} \end{array}$$



□ **GATE CS** – How many 256M X 8 RAM chips are required to build 32GB of RAM. Also Calculate the size of decoder.

$$\Rightarrow \text{Number of Chips} = \frac{\text{Total Size}}{\text{Available size}} = \frac{32 \text{ GB}}{256 \text{ m} \times 8}$$

$$= \frac{2^5 \times 2^{30} \times 2^3}{2^8 \times 2^{20} \times 2^3}$$

$$= 2^7$$

$$= \underline{128 \text{ Chips}}$$

3

I/p		O/p
7	X	128

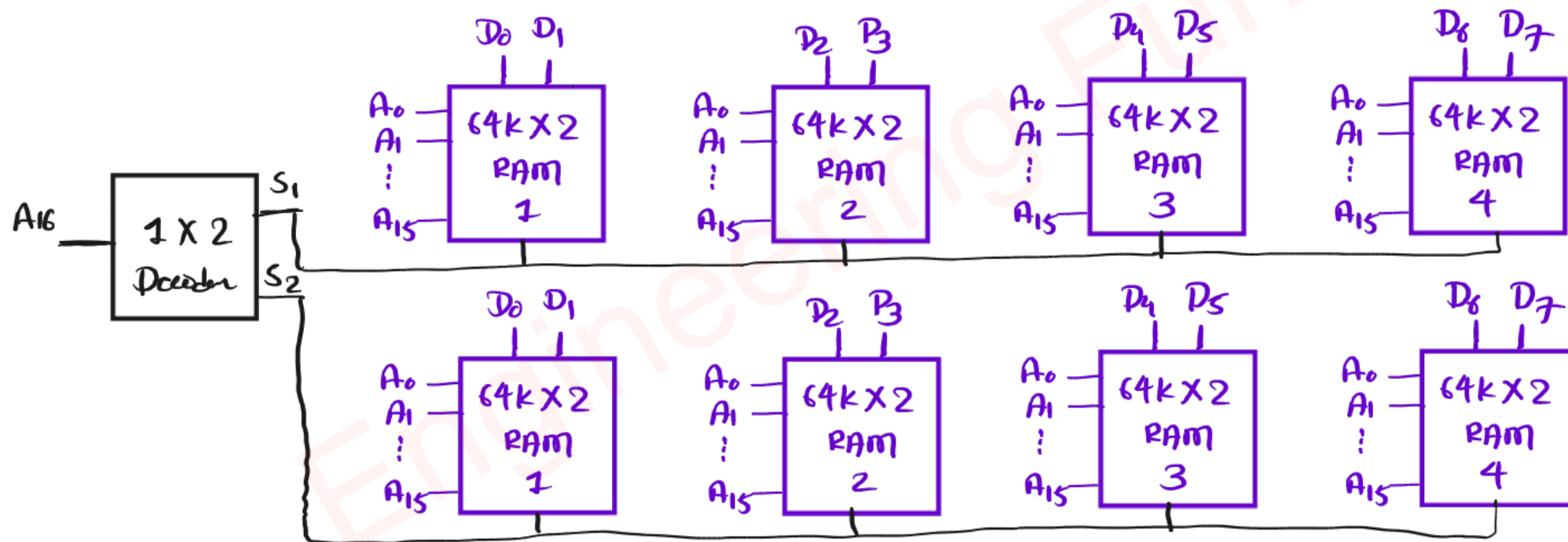
$\uparrow$ $\uparrow$
n 2^n

$\leftarrow 7 \times 128$

Memory Interfacing Questions of COA

- ❑ **GATE CS** – We are given with 64K X 2 RAM chips. How many chips required to build 128KB RAM. Also calculate the size of Decoder.

$$\Rightarrow \text{No of chips} = \frac{\text{Total Size}}{\text{Available Size}} = \frac{128 \text{ KB}}{64 \text{ K} \times 2} = \frac{2^7 \times 2^{10} \times 2^3}{2^6 \times 2^{10} \times 2^1} = 2^3 = \underline{8} \text{ chips}$$



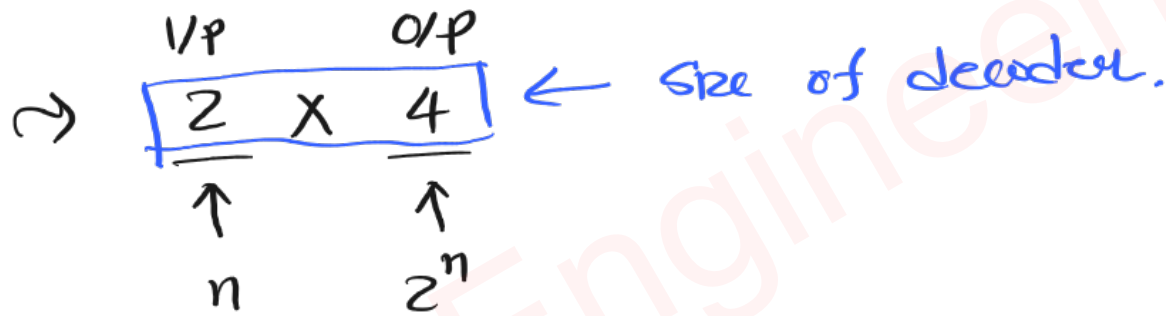
$$\Rightarrow \text{Decoder Output} = \text{No of chips} \times \frac{\text{Available data lines}}{\text{Req.}^{\text{ed}} \text{ data lines}} = 8 \times \frac{2}{8} = 2$$

□ **GATE CS** – We are given with 16K X 4 RAM chips. How many chips required to build 64KB RAM. Also calculate the size of Decoder.

$$\Rightarrow \text{No of chips} = \frac{\text{Total Size}}{\text{Available Size}} = \frac{64 \text{ KB}}{16\text{K} \times 4} = \frac{2^6 \times 2^{10} \times 2^3}{2^4 \times 2^{10} \times 2^2} = 2^3 = 8 \text{ chips}$$

$$\Rightarrow \text{Decoder Outputs} = \text{No of chips} \times \frac{\text{Available data lines}}{\text{Req.}^{\text{ed}} \text{ data lines}}$$

$$= 8 \times \frac{4}{8} = 4$$



Memory Interfacing Questions of COA

❑ **GATE 2009 CS / ISRO 2015 CS** – How many 32K X 1 RAM chips are needed to provide memory capacity of 256K bytes?

- A. 8
- B. 32
- ✓ C. 64
- D. 128

$$\Rightarrow \text{No of chips} = \frac{\text{Total Size}}{\text{Available Size}}$$

$$= \frac{256 \text{ KB}}{32 \text{ K} \times 1}$$

$$= \frac{2^8 \times \cancel{2^0} \times 2^3}{2^5 \times \cancel{2^0}}$$

$$= 2^6$$

$$= 64 \text{ chips}$$

☐ **GATE 2013 CS** – A RAM chip has capacity of 1024 words of eight bits each (1K X 8). The Number of 2 X 4 decoders with enable input line needed to construct a 16K X 16 from 1K X 8 RAM is

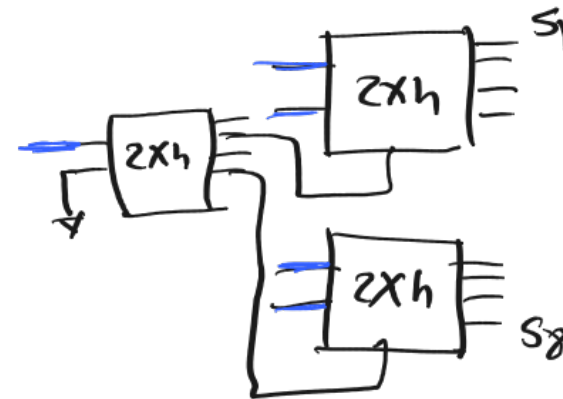
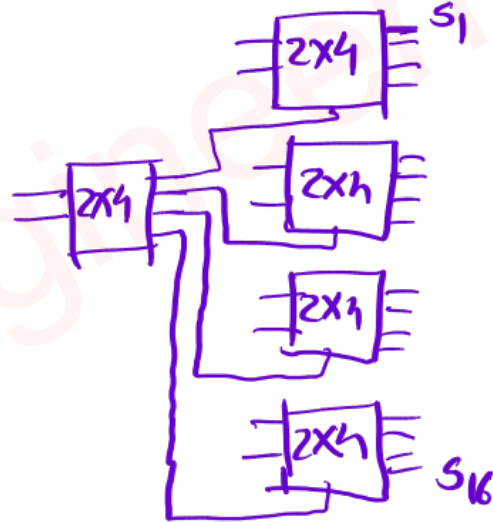
- A. 4
- ✓ B. 5
- C. 6
- D. 7

$$\Rightarrow \text{No of chips} = \frac{\text{Total Size}}{\text{Available Size}} = \frac{16K \times 16^2}{1K \times 8} = 32 \text{ chips}$$

$$\Rightarrow \text{No of Outputs with decoder} = \text{No of chips} \times \frac{\text{Available data lines}}{\text{Req.}^{\text{d}} \text{ data lines}}$$

$$= 32 \times \frac{8}{16} = 16$$

$$\Rightarrow \begin{array}{cc} \text{I/p} & \text{O/p} \\ \frac{4}{\uparrow} & \times \frac{16}{\uparrow} \\ n & 2^n \end{array}$$



$$\underline{3 \times 8} \leftarrow \underline{2 \times 4}$$

Registers of Basic Computer in COA

Memory 4K X 16	
Address	Data
⋮	⋮

IR – Instruction Reg.
[16 bits]

AC – Accumulator Reg.
[16 bits]

OUTR [8 bits]

I	Opcode	Operand
---	--------	---------

Basic Instruction Format

AR – Address Register
[12 bits]

PC – Program Counter
[12 bits]

DR – Data Register
[16 bits]

TR – Temporary Reg.
[16 bits]

INPR [8 bits]

- ❖ AC – Accumulator {16 bits for this case}
 - ❑ It is process register for ALU. ALU use it for all arithmetic and logical operations.
- ❖ TR – Temporary Register {16 bits for this case}
 - ❑ It holds data temporarily during instruction execution.
- ❖ OUTR & INPR – Input Output Registers {8 bits for this case}
 - ❑ OUTR holds data to be given to output devices.
 - ❑ INPR holds data given by input devices.

❖ Memory

- ❑ Here, memory is given by 4K X 16.
- ❑ Total Address of memory = $4K = 2^2 \times 2^{10} = 2^{12}$
- ❑ So, Numbers of Bits required to Access Memory is 12.
- ❑ Each Address has 16 bits of data, means word size is 16bits.
- ❑ Memory is used to store Instructions and data.

❖ Basic Instruction Format

- ❑ Here, basic instruction format is given with 16 bits.
- ❑ I bit = If it is 0 then direct addressing is used and If it is 1 then indirect addressing is used.
- ❑ Opcode = It explains that what operation should be performed by given instruction. {3 bits for this case}
- ❑ Operand = 12 bits of operand is used to take data from memory for given operation.

❖ AR/MAR – Address Register {12 bits for this case}

- ❑ It holds address of memory.

❖ PC – Program Counter {12 bits for this case}

- ❑ It holds address of next instruction in memory.

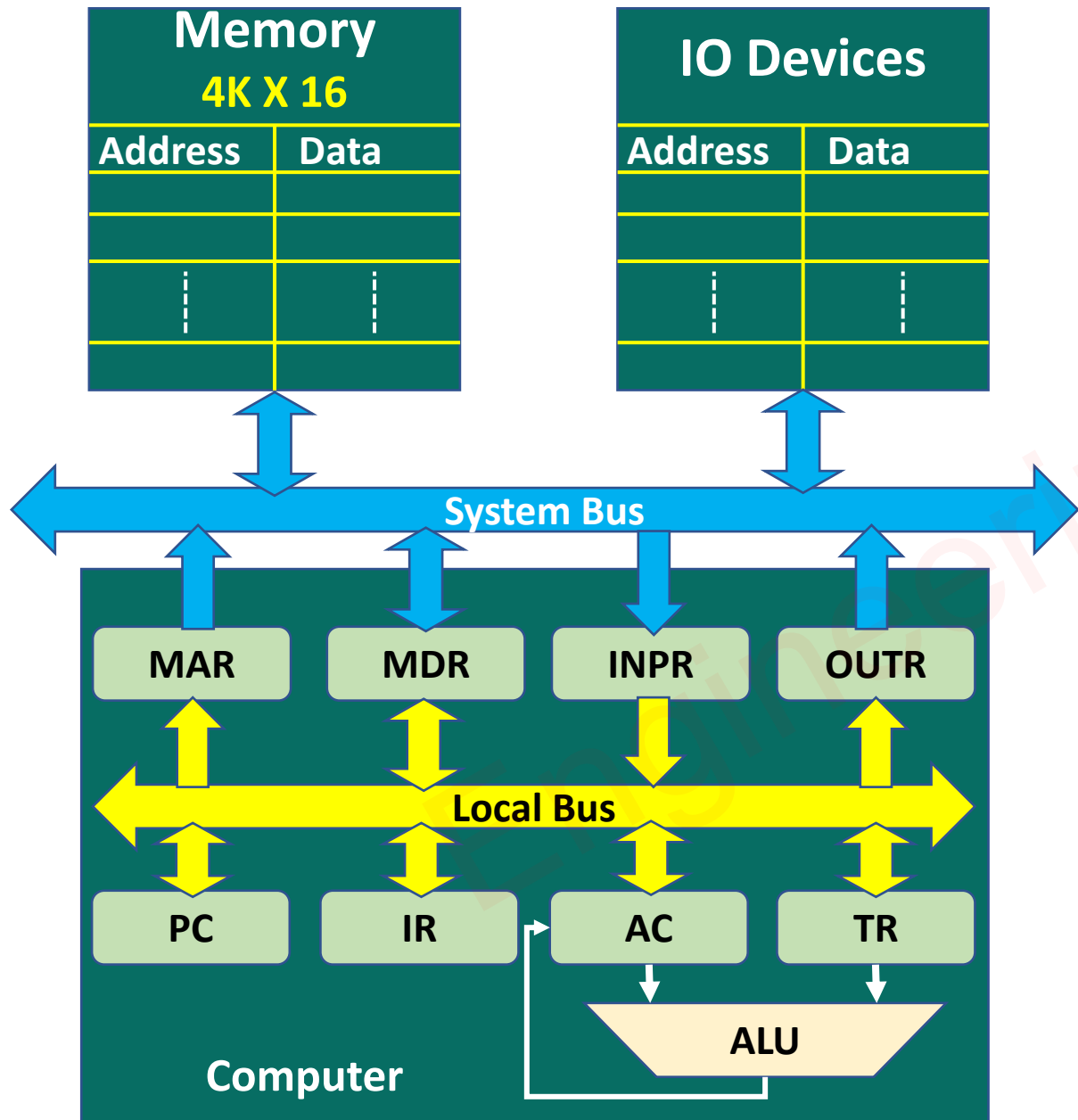
❖ IR – Instruction Register {16 bits for this case}

- ❑ After Fetching Instruction from memory, it is stored in Instruction Register IR.

❖ DR/MDR – Data Register {16 bits for this case}

- ❑ It has data operand read from the memory.

Registers Organization in COA



❖ Memory and IO Devices

- ❑ Memory is used to store Instructions and data of Programs.
- ❑ IO Devices are used to perform Input and Output operations.

❖ System Bus

- ❑ System Bus includes Address, Data and Control signals.
- ❑ Using system bus computer executes instructions, Takes data from input and gives data on to output.

❖ MAR – Memory Address Register & MDR – Memory Data Register

- ❑ MAR holds address of memory and MDR exchanges the data with respect to memory.

❖ INPR & OUTR

- ❑ It holds the data for Input & Output operations.

❖ Local Bus

- ❑ It provides internal communication with computer.

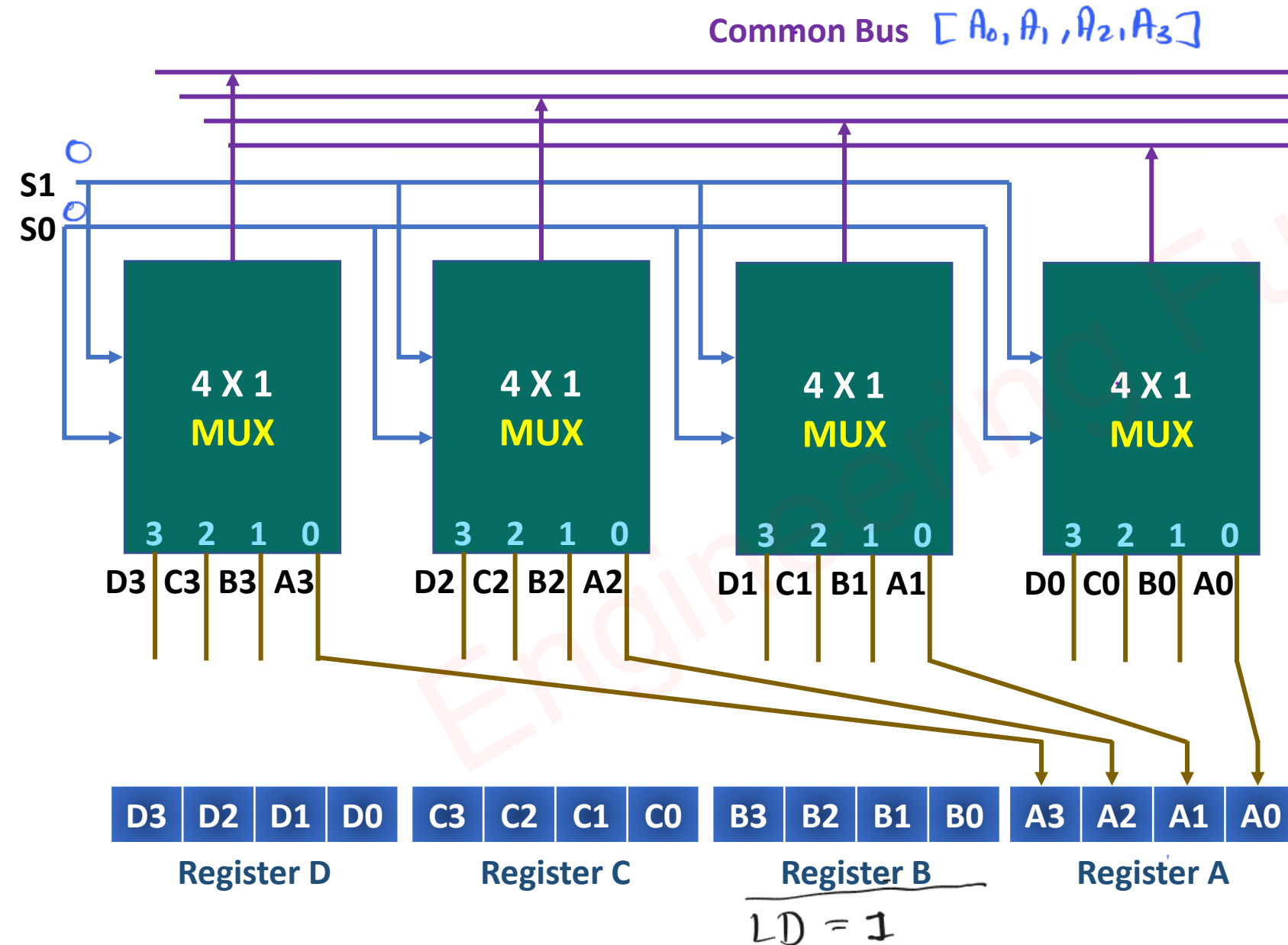
❖ Registers of Computer

- ❑ It is connected with local {Internal} bus of computer.

❖ ALU and Accumulator

- ❑ Most of Arithmetic and Logical task of ALU is performed with respect to Accumulator register only.

Common Bus Access using MUX in COA



→ $MOV\ B, A$

→ $LD = 1$ is given to B, by which now content of Common bus will come in B.

→ GATE Questions.

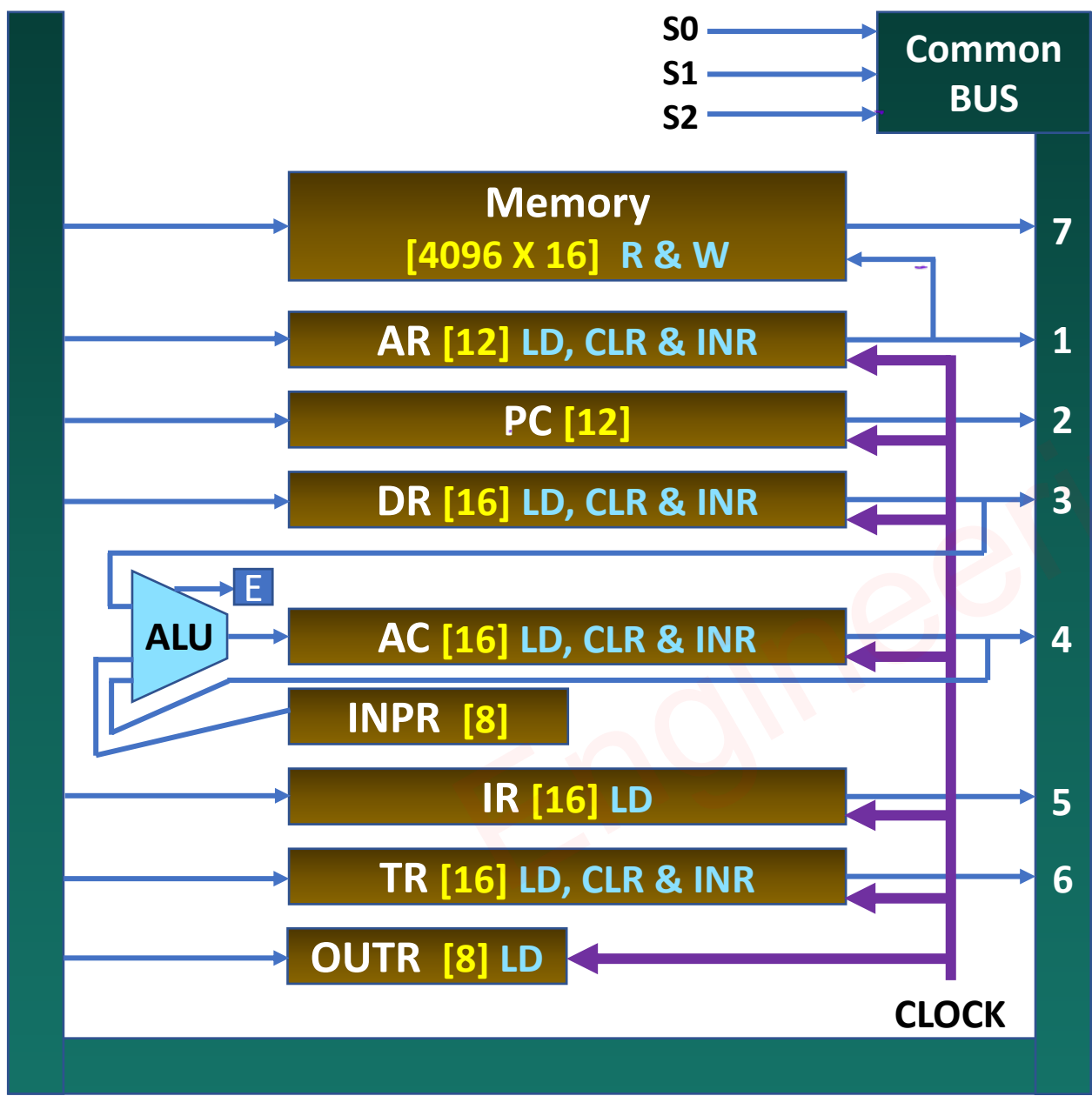
1) No of MUX = No of Bits in Reg.

2) Size of MUX = $[2^n \times 1]$

- No of Reg. = No of I/p

- n No of Selection lines.

Working of Computer using Common Bus in COA



⇒ Computer Structure

→ mux selection lines : It will select memory or any Register on common bus.

→ Common Bus : It has Add / data.

12 : Address lines.

16 : Data lines.

→ 16 lines, Add / data, are time multiplexed.

→ In case of Add, 1st four bits = 0.

→ This Computer will work w.r.t. Clock.

→ Registers have three control signals

- LD - Load

- CLR - Clear

- INR - Increment

→ Memory has two control signals

- R - Read

- W - Write.

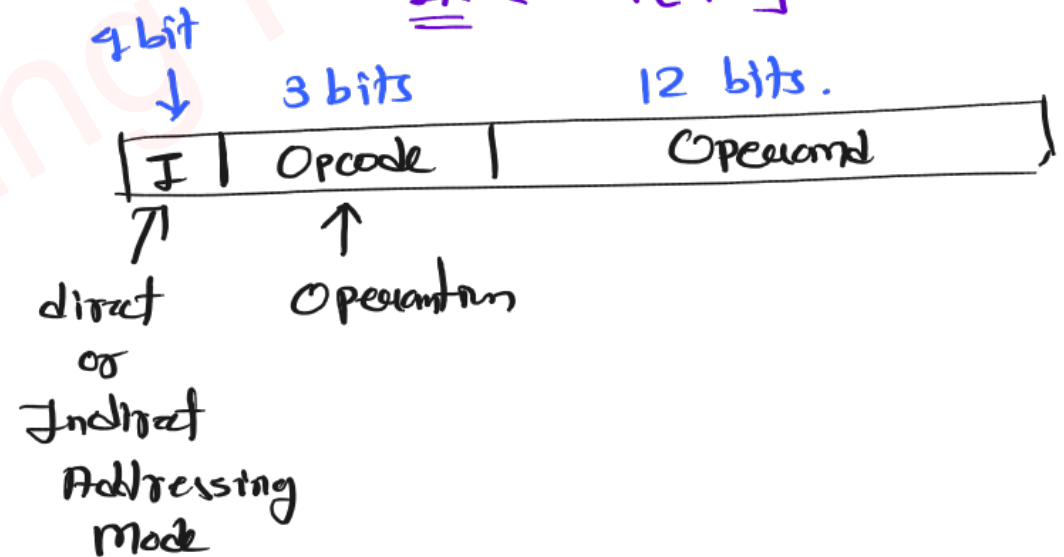
⇒ Instruction Execution

→ Address of next instruction is there with PC.

$$\underline{\underline{AR}} \leftarrow PC$$

→ Read control by CPU to memory.

$$\underline{\underline{IR}} \leftarrow m[AR]$$



Questions on Registers of Computer

❑ GATE 2020 CS – Consider the following data path diagram

❑ Execute : $R_0 \leftarrow R_1 + R_2$

❑ The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively

1. $R2_r, TEMP1_r, ALU_{ADD}, TEMP2_w$ ④

2. $R1_r, TEMP1_w$ ③

3. PC_r, MAR_w, MEM_r ①

4. $TEMP2_r, R0_w$ ⑤

5. MDR_r, IR_w ②

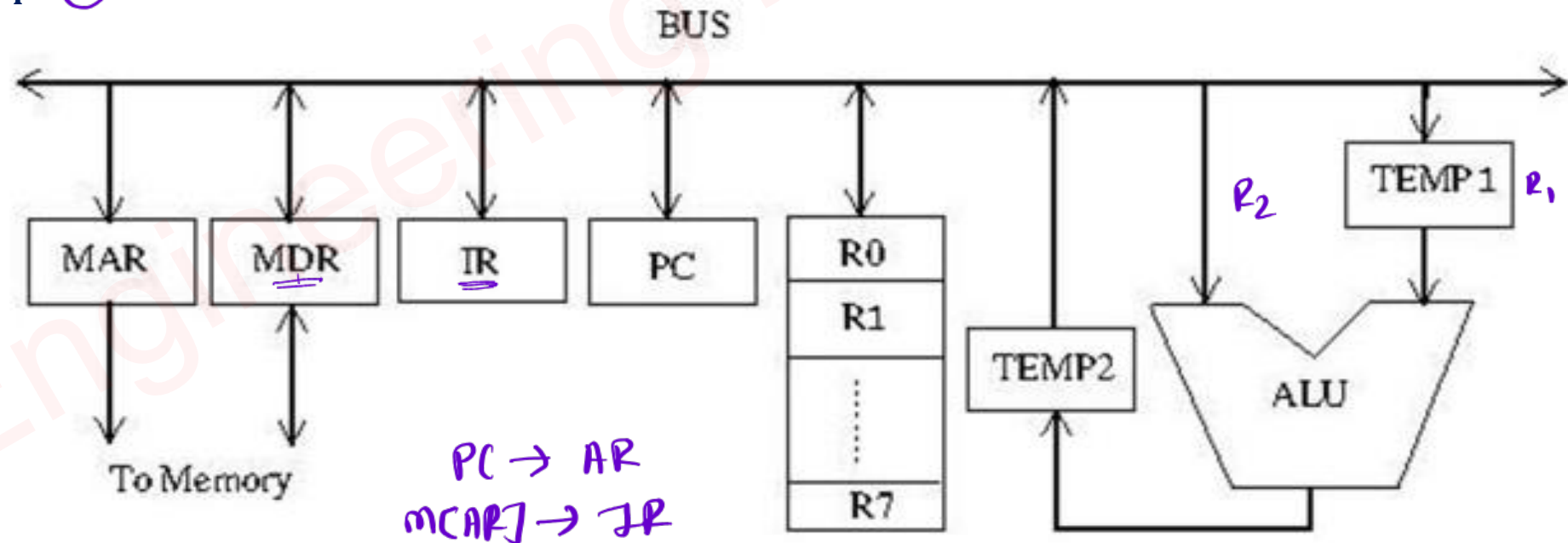
❑ Find the correct order:

A. 2 1 4 5 3 ✗

B. 1 2 4 3 5 ✗

✓ C. 3 5 2 1 4

D. 3 5 1 2 4



☐ **GATE CS** – Which of the following register is connected to system bus?

- A. PC
- ☒ B. MAR
- C. IR
- D. MDR

☐ **GATE CS** – Which of following register always points to the next instruction to be executed?

- ☒ A. PC
- B. MAR
- C. IR
- D. MDR

☐ **GATE CS** – Which of following register contains the most recently fetched instruction?

- A. PC
- B. MAR
- ☒ C. IR
- D. MDR

☐ **GATE CS** – Which of the following register contains the address of a main memory location from where instruction has to be fetched?

- A. PC
- ✓ B. MAR
- C. IR
- D. MDR

☐ **GATE CS** – Which of the following register contains the contents found at the address held by MAR

- A. PC
- B. AC
- C. IR
- ✓ D. MDR

☐ **ISRO 2008 CS** – The Memory Address Register

- A. Is a hardware memory device which denotes the location of the current instruction being executed
- B. Is a group of electrical circuit, that performs the intent of instructions fetched from memory
- C. Contains the address of memory location that is to be read from or store into
- D. Contains a copy of the designated memory location specified by the MAR after “READ” or the new contents of the memory prior to a “Write”